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CHROMATOGRAPHY

Separate Ink Colours

A Magical Science Experiment!

■ Subject	Science Kitchen Experiments
■ Duration	30 Minutes
■ Theme	Colour Detection Chromatography
■ Type	Hands-On Discovery Experiment
■ Context	Indian Classroom — CBSE / ICSE / State Board
■ Prepared by	Planrs Academy — Curriculum Team

This detailed lesson guide covers the complete chromatography experiment — from the science hook to hands-on steps, vocabulary, real-life connections, assessment, and extension activities. Every child leaves as a colour detective!

1 ■ The Big Question — Hook!

The session opens with one electrifying question that sparks curiosity:

Is Black Ink REALLY Black?

Or is it hiding secret colours inside? ■

■ Classroom Hook (First 3 Minutes)

- Teacher draws a thick black dot on filter paper — NO explanation yet.
- Dips the paper into a katori of water in front of the class.
- Students watch in silence as colours BEGIN to appear...
- Then: 'What did you see? What do YOU think happened?'
- Accept ALL guesses — wrong answers are BRAVE answers!

■ Colour Detective Mission

- Students are now COLOUR DETECTIVES.
- Their mission: discover what colours are hiding inside black ink.
- Their tools: filter paper, a sketch pen, and water.
- Their superpower: careful OBSERVATION.

2 ■ The Big Science Word!

CHROMATOGRAPHY — Breaking It Down

CHROMA	+	GRAPHY	=	CHROMATOGRAPHY
Greek for 'COLOUR'		Greek for 'WRITING'		Colour Writing! Separating Colours

■ Child-Friendly Definition

- Chromatography is a way scientists SEPARATE colours that are mixed together.
- It's like magic — finding hidden colours inside something that looks plain!
- The word comes from GREEK — ancient scientists were curious about colours too!

3 ■ What Do We Need?

Materials List — Indian Classroom Edition

#	Material	Indian Options / Notes	✓
1	Black sketch pen or marker	Camlin / Natraj / Reynolds ■ Must be WATER-BASED (washable). Oil-based permanent markers will NOT work!	■
2	Filter paper strip (2 cm × 10 cm)	Coffee filter (Bru/Nescafé) Blotting paper from stationery shop Newspaper or tissue (backup option)	■
3	Small bowl / katori	Steel katori, plastic cup, or cut plastic bottle bottom	■
4	Water	Plain tap water — do NOT add salt or soap for the basic experiment	■
5	Pencil	Wooden pencil — NOT a pen! Pencil lines won't mix with water	■
6	Ruler	To measure 2 cm from bottom and draw a straight start line	■

■ Teacher Preparation Checklist (Day Before)

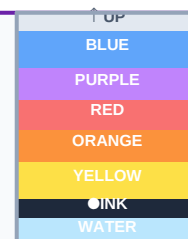
- Test the experiment yourself — different pen brands give different results!
- Pre-cut filter paper strips (2 cm × 10 cm) — one per student or pair.
- Fill katoris halfway with water just before the session begins.
- Check pens: press on a damp finger — if it smears, it's water-based. ✓
- Keep paper towels or old newspaper on desks to manage spills.
- Total cost of materials: under ■50 from any Indian kirana / stationery shop.

4 ■ Step-by-Step Experiment

Complete Procedure — 5 Steps

STEP 1 Prepare Your Paper

- Cut filter paper into a strip: 2 cm wide × 10 cm long.
- Ask a grown-up to help with scissors.
- Measure carefully using your ruler.
- This strip will be your MAGIC COLOUR PAPER!



STEP 2 Draw the Start Line

- Use a PENCIL to draw a line 2 cm from the bottom of your strip.
- Why pencil? Pencil marks won't mix with water like ink does!
- Place ruler at the bottom, measure 2 cm up, draw a straight line.

STEP 3 Make the Ink Dot

- Use your BLACK marker to make a thick dot ON the pencil line.
- Make the dot about 0.5 cm wide — the size of a small pea.
- Press firmly so the ink is nice and dark. More ink = more colours!

STEP 4 Add Water

- Pour water into your katori — about 1 cm deep.
- ■ VERY IMPORTANT: The water level must be BELOW the ink dot!
- If the dot touches water directly, colours won't separate properly.
- Gently place the paper strip in the katori — bottom touches water.

STEP 5 Watch the Magic!

- Hold the paper strip upright — DON'T let go!
- The water climbs UP the paper by itself — this is capillary action!
- Watch carefully — the colours begin to SEPARATE into bands!
- A rainbow of hidden colours appears! Count how many you see! ■

■ Safety Rules

- Do NOT drink the water — it contains ink dye.
- Handle scissors carefully — ask a grown-up for help.
- Keep water away from notebooks and bags.
- Wash hands after the experiment.

5 ■■ The Science Behind It!

Key Concept	What It Means	Indian Analogy
Black ink is a MIXTURE	Black ink contains many coloured dyes blended together. It looks black because all the colours are mixed at once.	Like mixing all your rangoli colours — you get a dark muddy colour!
Water as a carrier (Capillary Action)	Water travels UP the paper through tiny gaps between paper fibres, even against gravity. It carries the ink with it.	Like tea climbing up a cloth dipped in chai — water moves on its own!
Colours travel at different speeds	Each dye has a different weight. Lighter dyes travel HIGHER and FASTER. Heavier dyes stay LOWER and move SLOWLY.	Like fast runners vs slow walkers — they end at different places!
Separation	Because colours travel different distances, they separate into individual coloured bands on the paper.	Like using a chai strainer (chalni) — each part ends up in a different place!

■ Colours You Might See (depends on brand!)

- Blue • Purple • Violet • Red • Pink • Orange • Yellow • Green
- Different brands (Camlin, Natraj, Reynolds) give DIFFERENT colour combinations.
- 'Do all black pens have the same hidden colours?' — great extension question!

6 ■ Observe & Discuss

Guided Observation Questions (Teacher Asks)

Q#	Question for Students	Expected Response
Q1	What do you see on the paper strip?	Bands / stripes of colour
Q2	How many colours can you spot?	3–6 colours (varies by brand)
Q3	Which colour went the HIGHEST?	Blue or violet (lightest dye)
Q4	Which colour stayed the LOWEST?	Yellow or orange (heaviest dye)
Q5	Is black ink really just black?	No! It has many hidden colours!
Q6	What did the water DO to the ink?	Carried it up and separated the colours
Q7	What makes the water climb up by itself?	Allow any guess — introduce 'capillary action'

■ Student Observation Recording

- Ask students to COUNT the colours they see and write the number.
- DRAW their chromatography strip and colour in the bands they saw.
- WRITE: 'I was surprised that _____'
- COMPLETE: 'Black ink has _____ colours hiding inside.'

7 ■■ Science in Our Daily Life!

Indian Real-Life Connections

■ Rangoli Colours

The beautiful patterns in Diwali rangoli use many mixed colour powders. Mixing them all creates a dark colour — just like black ink! Chromatography would separate those colours back out.

■ Holi Gulal

The bright coloured powder thrown during Holi are mixtures of dyes! Scientists use chromatography to check if gulal colours are safe for skin.

■ Mithai & Food Safety

Colourful Indian sweets get their colours from food dyes. FSSAI (Food Safety Standards Authority of India) scientists use chromatography to check those colours are safe to eat!

■ Chai Strainer

When Mummy makes chai, she uses a strainer (chalni) to separate tea leaves from the liquid. Chromatography works similarly — water acts as the mover and separates mixed colours.

■ Dupatta / Sari Fabric

Have you noticed water climbing up a wet cloth by itself? This is CAPILLARY ACTION — the same force that pulls water up the filter paper in our experiment!

8 ■ Our Science Vocabulary!

Words We Learned Today

CHROMATOGRAPHY	→	A technique to separate colours mixed together
MIXTURE	→	Two or more things combined together
SEPARATION	→	Taking apart things that were mixed
OBSERVATION	→	Watching very carefully, like a detective
CAPILLARY ACTION	→	Water climbing up through tiny spaces on its own
DYE / PIGMENT	→	The coloured substance that gives ink its colour
SOLVENT	→	A liquid (like water) that dissolves and carries things

■ Vocabulary Practice Tips

- Say each word ALOUD 3 times as a class — clap the syllables!
- CHRO-MA-TO-GRA-PHY → 5 claps! This is a BIG word — celebrate it!
- Use in a sentence: 'I used chromatography to find hidden colours.'
- Write it in science notebooks — draw the chromatography strip beside it.

9 ■ Think Like Scientists! — Extension

'What If?' Questions for Curious Minds

Question	Predict	Result
What if we try a BLUE pen?	Will blue have the same hidden colours as black?	Blue pens often separate into cyan + violet + yellow.
What if we use SALT water?	Will salt change how far colours travel?	Salt water changes separation — colours travel differently!
What if we try DIFFERENT paper?	Tissue vs newspaper vs filter paper — which works best?	Thinner, more fibrous paper gives cleaner separation.
What if we compare PEN BRANDS ?	Do Camlin, Natraj, Reynolds pens have the same hidden colours?	Likely not! Great science fair project opportunity!

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■ 30-Minute Session Outline

Time	Phase	Activity	Teacher Action
0:00–3:00	■ HOOK	Magic Pen Moment	Silently dip ink paper in water. Let students gasp. Ask: 'What did you see?'
3:00–6:00	■ QUESTION	Introduce the Big Word	Ask: 'Is black ink really black?' Break down CHROMATOGRAPHY word by word.
6:00–8:00	■ MATERIALS	Materials & Safety Check	Show each material. Name Indian alternatives. State 4 safety rules clearly.
8:00–18:00	■ EXPERIMENT	Students Do the Experiment	Guide Steps 1–5. Circulate and ask observation questions. Let students discover!
18:00–22:00	■ OBSERVE	Share & Discuss	Students share what colours they saw. Teacher records results on the board.
22:00–26:00	■ CONNECT	'What If?' + Indian Links	Rangoli, chai, FSSAI, Holi gulal connections. Try 1 more colour if time allows.
26:00–29:00	■ RECORD	Science Notebook	Draw strip. Colour the bands. Complete fill-in sentences.
29:00–30:00	■ CELEBRATE	Exit Ticket + Shabash!	Q&A: Is black really black? What's the science word? Celebrate loudly!

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■ Assessment — Exit Ticket Quiz

5 Quick Questions — After the Experiment

Q #	Question	Type	Answer
Q1	Is black ink really black — or is it something else?	Spoken / Written	No! It's a MIXTURE of many colours.
Q2	What is the big science word for separating colours?	Fill-in-blank	CHROMATOGRAPHY
Q3	True or False: All black pens have the same hidden colours.	True / False	FALSE — different brands give different colours.
Q4	Draw a chromatography strip and label at least 2 colours you saw.	Draw & Label	Any 2 correct colours with labels.
Q5	What does water do to help in this experiment?	Open-ended	It carries the ink up the paper and separates the colours.

Observation Rubric (Teacher Use)

Score	What it looks like
3 ■■■	Noticed 3+ colours, described their position on the strip, asked at least 1 spontaneous question.
2 ■■	Noticed 2 colours, made a clear observation, participated actively in discussion.
1 ■	Participated in the experiment, made at least 1 observation (any kind).
0	Did not participate or could not make any observation — provide additional support.

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■ Differentiation Guide

■ Needs More Support	■ Ready for More	■ At Home / Hybrid
• Pre-draw the start line on their strip • Work in pairs — peer support helps • Use a wider strip (3 cm) for easier handling • Provide a word bank for the worksheet • Focus on observing — teacher scribes for them • Use picture-based answer options	• Try 3 different coloured markers • Compare pen brands (Camlin vs Natraj) • Try salt water vs plain water • Try tissue vs newspaper vs filter paper • Create a bar graph of colour heights • Research: Why do colours separate at all?	• Any pen + coffee filter + glass works • Share result photo with family • Ask: Which brand has the most colours? • Parent card with Try-at-Home steps • No chemicals needed — fully safe • Materials under ₹20 from home items

■ You Are Scientists Today! ■

Well Done! Wah Wah! Super! Shabash!

Keep asking WHY and WHAT IF!

✓ What students learned today

- ★ Black ink is a MIXTURE of many hidden colours
- ★ CHROMATOGRAPHY = separating colours using water and paper
- ★ Water travels up paper by CAPILLARY ACTION
- ★ Different colours travel at DIFFERENT SPEEDS → they SEPARATE
- ★ Scientists use this to keep our food SAFE — FSSAI!